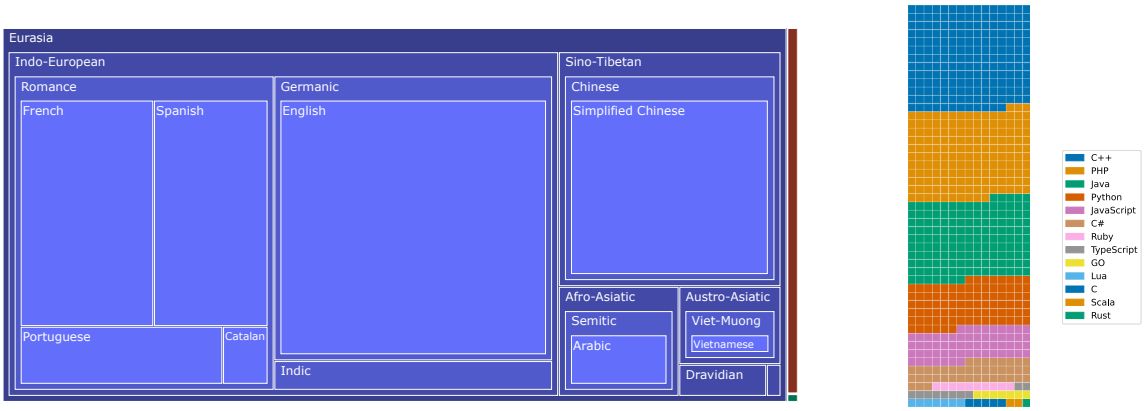

Figure 3: Graphical overview of the ROOTS corpus. **Left:** A treemap plot of the language families of all 46 natural languages where surface is proportional to the number of bytes. Indo-European and Sino-Tibetan families overwhelm the plot with a combined total of 1321.89 GB. The thin orange surface represents 18GB of Indonesian data and the green rectangle 0.4GB constituting the Niger-Congo language family subset. **Right:** A waffle plot of the distribution of the 13 programming languages by size, where one square represents approximately 200MB.

Deduplication and Privacy Redaction Finally, we removed near-duplicate documents with two deduplication steps and redacted Personal Identifiable Information (such as social security numbers) that we could identify from the OSCAR version of the corpus—as it was deemed to be the source that presented the highest privacy risks, prompting us to apply regex-based redaction even in cases where the expressions had some false positives.

3.1.4 Prompted Datasets

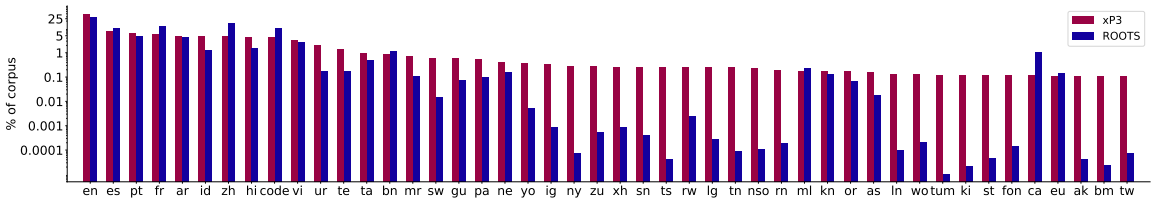

Figure 4: Language distribution of the prompted dataset xP3 closely follows ROOTS.

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Multitask prompted finetuning (also referred to as instruction tuning) involves finetuning a pretrained language model on a training mixture composed of a large set of different tasks specified through natural language prompts. T0 (Sanh et al., 2022) (developed as part of BigScience) demonstrated that language models finetuned on a multitask mixture of prompted datasets have strong zero-shot task generalization abilities. Moreover, T0 was shown to outperform language models that are an order of magnitude larger but did not undergo such finetuning. Motivated by these results, we explored using existing natural language datasets to carry out multitask prompted finetuning.

T0 was trained on a subset of the Public Pool of Prompts (P3), a collection of prompts for various existing and open-source English natural language datasets. This collection of prompts was created through a series of hackathons involving BigScience collaborators and where hackathon participants wrote a total of of 2000+ prompts for 170+ datasets. Datasets in P3 cover a variety of natural language task including sentiment analysis, question answering, and natural language inference and exclude harmful content or non-natural language such as programming languages. PromptSource (Bach et al., 2022),[16] an open-source toolkit (also developed as part of BigScience) facilitated creating, sharing and using natural language prompts. Full details of the collection process are given in (Sanh et al., 2022; Bach et al., 2022).

After pretraining BLOOM, we applied the same massively multitask finetuning recipe to equip BLOOM with multilingual zero-shot task generalization abilities. We refer to the resulting models as BLOOMZ. To train BLOOMZ, we extended P3 to include new datasets in languages other than English and new tasks, such as translation. This resulted in xP3, a collection of prompts for 83 datasets covering 46 languages and 16 tasks. As highlighted in Figure 4, xP3 mirrors the language distribution of ROOTS. Tasks in xP3 are both cross-lingual (e.g. translation) and monolingual (e.g. summarization, question answering). We used PromptSource to collect these prompts, adding additional metadata to the prompts, such as input and target languages. To study the importance of multilingual prompts, we also machine-translated English prompts in xP3 to the respective dataset languages to produce a collection called xP3mt. Further details on the prompt collection for xP3 and xP3mt are given in Muennighoff et al. (2022b).

3.2 Model Architecture

This section discusses our design methodology and the architecture of the BLOOM model. In-depth studies and experiments can be found in Le Scao et al. (2022) and Wang et al. (2022a). We first review our design methodology, then motivate our choice of training a causal decoder-only model. Finally, we justify the ways that our model architecture deviates from standard practice.

3.2.1 Design Methodology

The design space of possible architectures is immense, making exhaustive exploration impossible. One option would be to exactly replicate the architecture of an existing large language model. On the other hand, a great deal of work on improving existing architectures has seen relatively little adoption (Narang et al., 2021); adopting some of these recommended practices could yield a significantly better model. We take a middle ground and focus on model families that have been shown to scale well, and that have reasonable support in publicly available tools and codebases. We ablate components and hyperparameters of the models, seeking to make the best use of our final compute budget.

Experimental Design for Ablations One of the main draws of LLMs has been their ability to perform tasks in a "zero/few-shot" way: large enough models can perform novel tasks simply from in-context instructions and examples (Radford et al., 2019), without ded-

16. `github.com/bigscience-workshop/promptsource`

icated training on supervised samples. Accordingly, and because finetuning a 100B+ model is unwieldy, we focused our evaluation of architectural decisions on zero-shot generalization, and do not consider transfer learning. Specifically, we measured zero-shot performance on diverse aggregates of tasks: 29 tasks from the EleutherAI Language Model Evaluation Harness (EAI-Eval, Gao et al. (2021)), and 9 tasks from the evaluation set of T0 (T0-Eval, Sanh et al. (2022)). There is significant overlap between the two: only one task from T0-Eval (StoryCloze) is not in EAI-Eval, although all prompts between the two are different. See Le Scao et al. (2022) for a detailed list of tasks and baselines. We also note that our tasks aggregates share 17 of the 31 tasks of the evaluation of GPT-3 (Brown et al., 2020).

We conducted our ablation experiments using smaller models. We used the 6.7B parameter scale for the pretraining objective ablations (Wang et al., 2022a) and the 1.3B scale for the rest including position embeddings, activations, and layer normalization (Le Scao et al., 2022). Recently, Dettmers et al. (2022) identified a phase transition for models larger than 6.7B, in which the emergence of "outliers features" is observed. This questions whether results obtained at the 1.3B scale should be assumed to extrapolate to our final model size.

Out-of-scope Architectures We did not consider mixture-of-experts (MoE) (Shazeer et al., 2017), due to a lack of widely used GPU-based codebases suitable for training them at scale. Similarly, we also did not consider state-space models (Gu et al., 2020). At the time of the design of BLOOM, they consistently underperformed in natural language tasks (Gu et al., 2021). Both of these approaches are promising, and have now demonstrated competitive results–at large scales for MoE (Fedus et al., 2022; Srivastava et al., 2022), and at smaller scale for state-space models with H3 (Fu et al., 2023).

3.2.2 Architecture and Pretraining Objective

Although most modern language models are based on the Transformer architecture, there are significant deviations between architectural implementations. Notably, while the original Transformer is based on an encoder-decoder architecture, many popular models have opted for encoder-only (e.g. BERT, (Devlin et al., 2019)) or decoder-only (e.g. GPT, (Radford et al., 2018)) approaches. Currently, all state-of-the-art language models over 100 billion parameters are causal decoder-only models (Brown et al., 2020; Rae et al., 2021; Chowdhery et al., 2022). This is in opposition to the findings of Raffel et al. (2020), in which encoder-decoder models significantly outperform decoder-only models for transfer learning.

Prior to our work, the literature was lacking a systematic evaluation of the zero-shot generalization capabilities of different architectures and pretraining objectives. We explored this question in Wang et al. (2022a) where we evaluated encoder-decoder and decoder-only architectures and their interactions with causal, prefix, and masked language modeling pretraining objectives. Our results show that immediately after pretraining, causal decoder-only models performed best – validating the choice of state-of-the-art LLMs. Furthermore, they can be more efficiently adapted after pretraining to a non-causal architecture and objective–an approach which has been further explored and confirmed by Tay et al. (2022).

3.2.3 Modeling Details

Beyond choosing an architecture and pretraining objective, a number of changes to the original Transformer architecture have been proposed. For example, alternative positional